

Exploratory Data Analysis (EDA) Report

Project Title: Titanic Dataset Analysis

Introduction

In this project, I performed **Exploratory Data Analysis (EDA)** on the Titanic dataset. The main purpose of this analysis is to understand the dataset, clean the data, and identify important patterns before building a machine learning model. EDA helps us understand the characteristics of the data and improves the quality of the dataset.

Objective

The objectives of this project are:

- To understand the Titanic dataset.
- To check the structure of the dataset.
- To identify missing values and duplicate records.
- To calculate basic statistical measures.
- To visualize the data using graphs.
- To analyze passenger details such as age, sex, class, fare, and survival.

Dataset Description

First, I imported the required libraries such as **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **Scikit-learn**. Then I loaded the **Titanic dataset** using Pandas. I used functions like `head()`, `tail()`, `columns`, `shape`, `info()`, and `describe()` to understand the dataset. These functions helped me know the number of rows, columns, data types, and summary statistics.

Data Cleaning

After loading the dataset, I checked for missing values using `isnull().sum()`. I also checked for duplicate records using `duplicated()` and removed them using `drop_duplicates()`. Cleaning the dataset helps improve the accuracy of future analysis and machine learning models.

Statistical Analysis

I calculated the following statistical measures for the **Age** column:

- Mean
- Median
- Variance
- Standard Deviation

These measures helped me understand the average age of passengers and how the age values are distributed.

Data Analysis

To understand the dataset better, I performed some filtering and grouping operations:

- Displayed passengers whose **Age is greater than 30**.
- Displayed all **male passengers**.
- Displayed passengers who travelled in **First Class (Pclass = 1)**.
- Calculated the **average fare** for each passenger class using `groupby()`.
- Calculated the **survival rate** based on gender.

These operations helped me understand the relationship between passenger details and survival.

Data Visualization

I created the following graphs:

- **Bar Chart** to show the distribution of male and female passengers.
- **Line Chart** to observe the trend in gender distribution.
- **Histogram** to view the frequency distribution of passenger gender counts.

These visualizations made it easier to understand the dataset and compare different categories.

Observations

- The dataset was loaded successfully.

- Missing values and duplicate records were checked.
- The Age column was analyzed using statistical measures.
- Male and female passenger distributions were visualized.
- The average fare varied across passenger classes.
- The survival rate differed between male and female passengers.
- Filtering and grouping operations helped in understanding passenger information more clearly.

Conclusion

This project helped me understand how to perform Exploratory Data Analysis on the Titanic dataset. I learned how to inspect the dataset, clean the data, calculate statistical measures, perform filtering and grouping, and create visualizations. Through this analysis, I gained a better understanding of the passenger information and survival patterns. The cleaned dataset is now ready for further machine learning model development.